Ex4: Information Retrieval using NLTK

Dataset: Amazon Fine Food Reviews (Kaggle) – `Reviews.csv`

The dataset (~300 MB) is downloaded automatically from Kaggle the first time you run the notebook. If you already have `Reviews.csv`, upload it to the Colab **Files** panel and it is used instead.

Task 1: Load the Amazon Fine Food Reviews dataset into a Pandas DataFrame

```
import os
import pandas as pd

file_name = "Reviews.csv"
if not os.path.exists(file_name):  # file not uploaded -> download from Kaggle
    import kagglehub
    path = kagglehub.dataset_download("snap/amazon-fine-food-reviews")
    file_name = os.path.join(path, "Reviews.csv")

df = pd.read_csv(file_name)
print("Shape of the dataset:", df.shape)
display(df.head())
```

```
Shape of the dataset: (568454, 10)
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Summary
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	1	5	1303862400	Good Quality Dog Food
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0	0	1	1346976000	Not as Advertised
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1	1	4	1219017600	"Delight" says it all
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3	3	2	1307923200	Cough Medicine
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0	0	5	1350777600	Great taffy

	Text
0	I have bought several of the Vitality canned d...
1	Product arrived labeled as Jumbo Salted Peanut...
2	This is a confection that has been around a fe...
3	If you are looking for the secret ingredient i...
4	Great taffy at a great price. There was a wid...

Task 2: Install and import the required Python libraries

```
!pip install -q nltk scikit-learn

import re
import numpy as np
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

pd.set_option('display.max_colwidth', 200)
```

Task 3: Import the required NLTK resources

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

```
[nltk_data]   Package punkt is already up-to-date!
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data]   Package stopwords is already up-to-date!
True
```

Task 4: Clean the dataset

Select the `Text` column, remove missing values, remove duplicate reviews, keep the first 10,000 reviews.

```
reviews = df[['Text']].copy()

print("Missing values in Text:", reviews['Text'].isnull().sum())
reviews = reviews.dropna(subset=['Text'])

print("Duplicate reviews:", reviews.duplicated(subset='Text').sum())
reviews = reviews.drop_duplicates(subset='Text')

reviews = reviews.head(10000).reset_index(drop=True)
print("Shape after cleaning:", reviews.shape)
display(reviews.head())
```

```
Missing values in Text: 0
Duplicate reviews: 174875
Shape after cleaning: (10000, 1)
```

	Text
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labr...
1	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".
2	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...
3	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (which was good) and made some cherry soda. The fl...
4	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.

Text Preprocessing

Task 5: Preprocessing function

Lowercase → remove punctuation, numbers and special characters → tokenize → remove English stopwords → join tokens.

```
stop_words = set(stopwords.words('english'))

def preprocess_text(text):
    text = str(text).lower()                                  # 1. convert to lowercase
    text = re.sub(r'<.*?>', ' ', text)                        #    remove HTML tags like <br />
    text = re.sub(r'[^a-z\s]', ' ', text)                     # 2. remove punctuation, numbers, special characters
    tokens = word_tokenize(text)                              # 3. tokenize into words
    tokens = [word for word in tokens if word not in stop_words]   # 4. remove English stopwords
    return ' '.join(tokens)                                   # 5. join tokens into a single string

print(preprocess_text("This is a GREAT product!!! Shipped in 2 days :) <br />Would buy again."))
```

```
great product shipped days would buy
```

Task 6: Apply the preprocessing function and store the result in `Cleaned_Text`

```
reviews['Cleaned_Text'] = reviews['Text'].apply(preprocess_text)
display(reviews.head())
```

	Text	Cleaned_Text
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labr...	bought several vitality canned dog food products found good quality product looks like stew processed meat smells better labrador finicky appreciates product better
1	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".	product arrived labeled jumbo salted peanuts peanuts actually small sized unsalted sure error vendor intended represent product jumbo
2	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	confection around centuries light pillowy citrus gelatin nuts case filberts cut tiny squares liberally coated powdered sugar tiny mouthful heaven chewy flavorful highly recommend yummy treat famil...
3	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (which was good) and made some cherry soda. The fl...	looking secret ingredient robitussin believe found got addition root beer extract ordered good made cherry soda flavor medicinal
4	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.	great taffy great price wide assortment yummy taffy delivery quick taffy lover deal

Task 7: Original and cleaned versions of five reviews

```
for i in range(5):
    print(f"Review {i + 1}")
    print("Original:", reviews.loc[i, 'Text'])
```

```
    print("Cleaned :", reviews.loc[i, 'Cleaned_Text'])
    print('-' * 120)
```

```
Review 1
Original: I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product
looks more like a stew than a processed meat and it smells better. My Labrador is finicky and she appreciates this product better
than  most.
Cleaned : bought several vitality canned dog food products found good quality product looks like stew processed meat smells better
labrador finicky appreciates product better
------------------------------------------------------------------------------------------------------------------------
Review 2
Original: Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an
error or if the vendor intended to represent the product as "Jumbo".
Cleaned : product arrived labeled jumbo salted peanuts peanuts actually small sized unsalted sure error vendor intended represent
product jumbo
------------------------------------------------------------------------------------------------------------------------
Review 3
Original: This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in this case
Filberts. And it is cut into tiny squares and then liberally coated with powdered sugar.  And it is a tiny mouthful of heaven.  Not
too chewy, and very flavorful.  I highly recommend this yummy treat.  If you are familiar with the story of C.S. Lewis' "The Lion,
The Witch, and The Wardrobe" - this is the treat that seduces Edmund into selling out his Brother and Sisters to the Witch.
Cleaned : confection around centuries light pillowy citrus gelatin nuts case filberts cut tiny squares liberally coated powdered
sugar tiny mouthful heaven chewy flavorful highly recommend yummy treat familiar story c lewis lion witch wardrobe treat seduces
edmund selling brother sisters witch
------------------------------------------------------------------------------------------------------------------------
Review 4
Original: If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in addition to the Root
Beer Extract I ordered (which was good) and made some cherry soda.  The flavor is very medicinal.
Cleaned : looking secret ingredient robitussin believe found got addition root beer extract ordered good made cherry soda flavor
medicinal
------------------------------------------------------------------------------------------------------------------------
Review 5
Original: Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.  If your a taffy
lover, this is a deal.
Cleaned : great taffy great price wide assortment yummy taffy delivery quick taffy lover deal
------------------------------------------------------------------------------------------------------------------------
```

Task 8: Convert the cleaned text into TF-IDF feature vectors

```
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(reviews['Cleaned_Text'])

print("Shape of the TF-IDF matrix:", tfidf_matrix.shape)
print("Total number of unique terms (vocabulary size):", len(tfidf.vocabulary_))
print("Sample terms:", tfidf.get_feature_names_out()[:20].tolist())
```

```
Shape of the TF-IDF matrix: (10000, 18328)
Total number of unique terms (vocabulary size): 18328
Sample terms: ['aa', 'aaaa', 'aacute', 'aadmit', 'aahhhs', 'ab', 'aback', 'abandon', 'abates', 'abberline', 'abbott', 'abby',
'abdominal', 'abiding', 'ability', 'able', 'ablution', 'abnormalities', 'abnormally', 'aboiut']
```

Information Retrieval

Task 10: Document retrieval function

Accepts a query → applies the same preprocessing → converts it to a TF-IDF vector → computes cosine similarity with every review → returns the Top 5 most similar reviews with similarity score, original review and cleaned review.

```
def retrieve_reviews(query, top_n=5):
    cleaned_query = preprocess_text(query)
    query_vector = tfidf.transform([cleaned_query])
    similarity = cosine_similarity(query_vector, tfidf_matrix).flatten()
    top_idx = similarity.argsort()[::-1][:top_n]

    results = pd.DataFrame({
        'Rank': range(1, top_n + 1),
        'Similarity Score': similarity[top_idx].round(4),
        'Original Review': reviews.loc[top_idx, 'Text'].values,
        'Cleaned Review': reviews.loc[top_idx, 'Cleaned_Text'].values,
    })

    print(f"Query        : {query}")
    print(f"Cleaned query: {cleaned_query}")
    print('=' * 120)
    for _, row in results.iterrows():
        print(f"Rank {row['Rank']} | Similarity Score: {row['Similarity Score']}")
        print("Original Review:", row['Original Review'])
        print("Cleaned Review :", row['Cleaned Review'])
        print('-' * 120)
    return results
```

Query Evaluation

Task 11: Test the retrieval system with the sample queries

```
queries = ["great product with fast shipping", "disappointed", "excellent taste", "poor packaging"]
all_results = {}
```

```
all_results[queries[0]] = retrieve_reviews(queries[0])
```

```
Query        : great product with fast shipping
Cleaned query: great product fast shipping
================================================================================================================================
Rank 1 | Similarity Score: 0.512
Original Review: Enjoyed the product and they also provided very fast shipping.  I'm about out and need to order more.
Cleaned Review : enjoyed product also provided fast shipping need order
--------------------------------------------------------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.5026
Original Review: The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.<br />I recommend this product.
Cleaned Review : tea good fresh enjoy shipping fast cost reasonable recommend product
--------------------------------------------------------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.4008
Original Review: The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping then both
products together, so not a great deal.
Cleaned Review : energy drink great product shipping price crazy tho paid shipping products together great deal
--------------------------------------------------------------------------------------------------------------------------------
Rank 4 | Similarity Score: 0.3987
Original Review: These are very good, they were a great price and fast-free shipping when you purchase enough food.  The only thing
is that the box was smaller than I guessed it would be.
Cleaned Review : good great price fast free shipping purchase enough food thing box smaller guessed would
--------------------------------------------------------------------------------------------------------------------------------
Rank 5 | Similarity Score: 0.3966
Original Review: Was exactly what I was looking for. Super fast shipping! A bit on the spicy side,but great taste!!!!<br />Would
definately purchase them again!
Cleaned Review : exactly looking super fast shipping bit spicy side great taste would definately purchase
--------------------------------------------------------------------------------------------------------------------------------
```

```
all_results[queries[1]] = retrieve_reviews(queries[1])
```

```
Query        : disappointed
Cleaned query: disappointed
================================================================================================================================
Rank 1 | Similarity Score: 0.4846
Original Review: I am a bit disappointed.  The flavor was not what I wanted or expected.
Cleaned Review : bit disappointed flavor wanted expected
--------------------------------------------------------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.3577
Original Review: The product is very good. Way too expensive and at almost 5.00 a box to get pancakes or biscuits I was disappointed
Cleaned Review : product good way expensive almost box get pancakes biscuits disappointed
--------------------------------------------------------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.3259
Original Review: I was disappointed in this product because I thought it would be bigger.<br />Also, it did not come with enough
icing. I had to use my own.
Cleaned Review : disappointed product thought would bigger also come enough icing use
--------------------------------------------------------------------------------------------------------------------------------
Rank 4 | Similarity Score: 0.3195
Original Review: Disappointed.  The big boxes had a very different flavor than the smaller boxes.  I am not sure why but not good.
Cleaned Review : disappointed big boxes different flavor smaller boxes sure good
--------------------------------------------------------------------------------------------------------------------------------
Rank 5 | Similarity Score: 0.3117
Original Review: Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at all.  Very
disappointed
Cleaned Review : plain nasty item tasted like cardboard watered flavor disappointed
--------------------------------------------------------------------------------------------------------------------------------
```

```
all_results[queries[2]] = retrieve_reviews(queries[2])
```

```
Query        : excellent taste
Cleaned query: excellent taste
================================================================================================================================
Rank 1 | Similarity Score: 0.5282
Original Review: It is very very good sweets<br />Amazon always is excellent<br />Amazon always is excellent<br />Amazon always is
excellent<br />Amazon always is excellent
Cleaned Review : good sweets amazon always excellent amazon always excellent amazon always excellent amazon always excellent
--------------------------------------------------------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.4605
Original Review: The Popchips were excellent - had excellent flavor and low in calories.  When you eat these - don't need another
potato chip!
Cleaned Review : popchips excellent excellent flavor low calories eat need another potato chip
--------------------------------------------------------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.4052
Original Review: Excellent product. Exactly what they said it would be. Superior chamomile from what I have ordered before from
other companies. Excellent value for the price.
Cleaned Review : excellent product exactly said would superior chamomile ordered companies excellent value price
--------------------------------------------------------------------------------------------------------------------------------
Rank 4 | Similarity Score: 0.3854
Original Review: We recieved the product on time and in excellent shape. It tastes good and meets our expectations. Thanks for the
excellent service.
Cleaned Review : recieved product time excellent shape tastes good meets expectations thanks excellent service
```

```
--------------------------------------------------------------------------------------------------------------------
Rank 5 | Similarity Score: 0.3832
Original Review: Title says it all, a Excellent quality product and Excellent service at a fair price. Product came ahead of time
and as expected.
Cleaned Review : title says excellent quality product excellent service fair price product came ahead time expected
--------------------------------------------------------------------------------------------------------------------
```

```
all_results[queries[3]] = retrieve_reviews(queries[3])
```

```
Query        : poor packaging
Cleaned query: poor packaging
====================================================================================================================
Rank 1 | Similarity Score: 0.3245
Original Review: I was delighted to find this product online, because I cannot always find it in the stores.  However, each package
I have opened thus far is at least half reduced to crumbs because of poor packaging for shipping.  I doubt that I will order more
using this method.
Cleaned Review : delighted find product online always find stores however package opened thus far least half reduced crumbs poor
packaging shipping doubt order using method
--------------------------------------------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.3085
Original Review: This review is for the packaging and not the actual mix, since the mix is a very subjective thing and my taste my
not match others and, some wrote about how poor the packaging was. All I can say is with mine there is more than enough labeling, on
the bags themselves and on the box there are instructions on how to. Granted the box is nothing heavy duty or special and the mix is
in plain white bags, there are more than enough instructions to go by.
Cleaned Review : review packaging actual mix since mix subjective thing taste match others wrote poor packaging say mine enough
labeling bags box instructions granted box nothing heavy duty special mix plain white bags enough instructions go
--------------------------------------------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.3082
Original Review: The soup is rich and tasty, but DON'T buy it from Amazon.  The packaging is so poor that the cans arive dented as
did the replacement shipment. I got my money back but now have to pay more to buy this wonderful soup from my local grocer.
Cleaned Review : soup rich tasty buy amazon packaging poor cans arive dented replacement shipment got money back pay buy wonderful
soup local grocer
--------------------------------------------------------------------------------------------------------------------
Rank 4 | Similarity Score: 0.292
Original Review: Didnt really enjoy these after wanting to try something other than aunt jemima's from Dominicks. The pancakes
tasted strange and sometimes forgettable, but it might have been due to the poor packaging they came in along with the syrup.
Cleaned Review : didnt really enjoy wanting try something aunt jemima dominicks pancakes tasted strange sometimes forgettable might
due poor packaging came along syrup
--------------------------------------------------------------------------------------------------------------------
Rank 5 | Similarity Score: 0.2804
Original Review: was really dissappointed in product... too salty and poor flavor....Convenience is what makes product worth
purchasing if you like pork rinds...
Cleaned Review : really dissappointed product salty poor flavor convenience makes product worth purchasing like pork rinds
--------------------------------------------------------------------------------------------------------------------
```

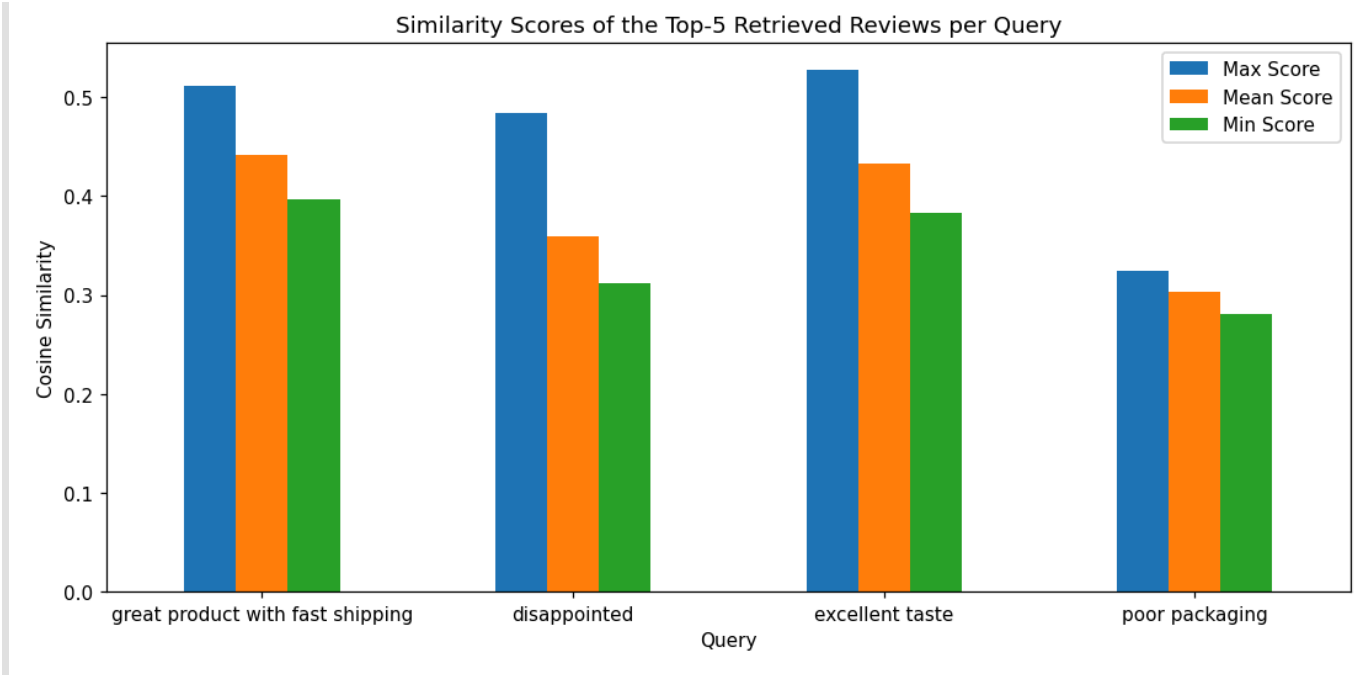
Task 12: Compare the retrieved reviews for different queries

```
summary = []
for query, res in all_results.items():
    terms = preprocess_text(query).split()
    any_term = res['Cleaned Review'].apply(lambda r: any(t in r.split() for t in terms)).mean() * 100
    all_terms = res['Cleaned Review'].apply(lambda r: all(t in r.split() for t in terms)).mean() * 100
    summary.append([query, ' '.join(terms), res['Similarity Score'].max(), res['Similarity Score'].mean(),
                    res['Similarity Score'].min(), any_term, all_terms])

summary_df = pd.DataFrame(summary, columns=['Query', 'Cleaned Query', 'Max Score', 'Mean Score', 'Min Score',
                                            '% Top-5 with any query term', '% Top-5 with all query terms'])
display(summary_df.round(4))

import matplotlib.pyplot as plt
summary_df.set_index('Query')[['Max Score', 'Mean Score', 'Min Score']].plot(kind='bar', figsize=(10, 5), rot=0)
plt.title('Similarity Scores of the Top-5 Retrieved Reviews per Query')
plt.ylabel('Cosine Similarity')
plt.tight_layout()
plt.show()
```

	Query	Cleaned Query	Max Score	Mean Score	Min Score	% Top-5 with any query term	% Top-5 with all query terms
0	great product with fast shipping	great product fast shipping	0.5120	0.4421	0.3966	100.0	0.0
1	disappointed	disappointed	0.4846	0.3599	0.3117	100.0	100.0
2	excellent taste	excellent taste	0.5282	0.4325	0.3832	100.0	0.0
3	poor packaging	poor packaging	0.3245	0.3027	0.2804	100.0	80.0



Comments on relevance

- **"great product with fast shipping"** – after preprocessing the query becomes *great product fast shipping* ("with" is a stopword). All five results contain at least one query word and four of them praise fast shipping, so they are relevant. No review contains all four words, so the best score is only 0.51. Rank 3 is only partly relevant: it calls the product great but complains about the shipping *price*.
- **"disappointed"** – every result contains the word "disappointed" and all are negative or mixed reviews, so the results are highly relevant. Short reviews rank highest because TF-IDF vectors are length-normalised, so one matching word carries more weight in a short review.
- **"excellent taste"** – gives the highest top score (0.53), but none of the results contain both words: they match "excellent" only (one review says "tastes", which does not match "taste" because no stemming/lemmatization is applied). The results are positive reviews, but not all of them are about taste, and Rank 1 is a repetitive review.
- **"poor packaging"** – has the lowest scores (0.28 – 0.32) because few reviews use these exact words. Four of the five results contain both words and most describe damaged or badly packed products, so the results are relevant.
- **Overall:** TF-IDF + cosine similarity retrieves reviews that share the query words and works well for distinctive keywords. It does not understand meaning or synonyms ("quick delivery" will not match "fast shipping"), it cannot match word forms ("tastes" vs "taste"), removing stopwords such as "not" loses negation, and short reviews tend to get higher scores.